

Azra Sakhawat

PhD (BIOCHEMIST)

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SUMMARY

Dynamic biochemist with a robust background in human recombinant protein production, purification, and bioassays. Expertise spans downstream processing, pharmacokinetics, and toxicity studies, including high impact projects like Semaglutide. I am a highly motivated researcher with a passion for cutting-edge research and a strong foundation in **Biochemistry, molecular biology, microbiology, therapeutic Exosomes projects, protein projects, Bioinformatics, R-language and Python programming** continuously striving to expand my expertise and contribute to innovative scientific solutions. I am aspiring to contribute to groundbreaking research as an outstanding professional in a challenging and dynamic environment where I can nurture and boost my skills to the utmost by complementing technical knowledge with practical experience. An active contributor to HEC-funded research, with a proven track record in quality control, technology transfer, and advancing human recombinant therapeutic production. Passionate about pushing the boundaries of biochemistry to deliver impactful scientific solutions.

PROFESSIONAL EXPERIENCE

Centre of Excellence in Molecular Biology – Lahore, Pakistan

City: Lahore | **Country:** Pakistan | **Name of unit or department:** Biopharmaceutical & Proteomics Lab (BPL)

Research Associate

[1 Jun 2024 – Current]

- Design, optimize, and execute experiments involving **recombinant DNA technology, PCR, RT-PCR, ELISA, HPLC, FPLC, and mammalian cell culture systems**.
- Lead projects on **biosimilars and therapeutic proteins** (e.g., Semaglutide, Filgrastim, IL-11), including **fermentation, Upstream & downstream processing, and quality analysis**.
- Conduct **molecular biology and biochemistry research**, including **protein expression, purification, and characterization**.
- Proficient in designing and executing **in vitro and in vivo bioassays**, pharmacokinetics (PK), and **toxicity studies** to evaluate efficacy and safety profiles of biologics.
- Development of Antibodies in non-Clinical Species (mice, rats and rabbits).
- Develop and validate indigenous PCR reagents and diagnostic kits in compliance with ➤ biosafety, ISO 90001:2015 standards.
- Hands-on experience in **exosome-based therapeutics**, including isolation, characterization, and drug/peptide loading strategies for targeted delivery applications.
- Strong foundation in **molecular biology, microbiology, and biochemistry techniques**, supporting translational and interdisciplinary research projects.

- Skilled in **bioinformatics analysis**, molecular modelling, and **computational biology tools**, with practical experience in **R and Python programming** for data analysis and visualization.
- Experienced in integrating **in silico, in vitro, and experimental data** to support hypothesis-driven research and therapeutic development.
- Demonstrated ability to work independently and collaboratively in **fast-paced research environments**, with a strong commitment to scientific rigor, innovation, and continuous learning.
- Supervised laboratory staff and guided students in molecular biology research at Production Unit of Biopharmaceutical Labs (BPL), CEMB

Teaching Experience

City: Lahore | **Country:** Pakistan | **Name of College:** Superior College, **Lecturer / Visiting Faculty** [2018 – 2019]

Oxford college (Affiliated with University of Sargodha) [2020 – 2023]

- Undergraduate and Graduate Teaching
- Theory & Practical Laboratory Instruction
- Student Mentorship & Academic Supervision
- Research-Oriented Teaching Approach
- Curriculum Delivery & Assessment

Internship Experience

City: Lahore | **Country:** Pakistan | **Name of Lab:** Shaikh Zayed Hospital [2016]

PUBLICATIONS

1. [2025]

Sakhawat A, Khan MU, Khan S, Ahmed N, Alhegaili AS, Alotaibi BS. **PD-L1 targeting in triple negative breast cancer:** in silico and in vitro validation of wasp venom peptide MP-1. Medical Oncology. 2025 Nov 24;43(1):14.

2. [2024]

Akram A, Sakhawat A, Ghani MU, Khan MU, Rehman R, Ali Q, Jin-Liang P, Ali D. Silibinins and curcumin as promising ligands against **mutant cystic fibrosis transmembrane regulator protein**. AMB Express. 2024 Jul 23;14(1):84.

3. [2024]

Khan MU, Sakhawat A, Rehman R, Wali AH, Ghani MU, Akram A, Javed MA, Ali Q, Yu-Ming Z, Ali D, Yu-ming Z. Identification of novel natural compounds against CFTR p. **Gly628Arg pathogenic variant**. AMB Express. 2024 Sep 9;14(1):99.

4. [2023]

Sakhawat A, Khan MU, Rehman R, Khan S, Shan MA, Batool A, Javed MA, Ali Q. Natural compound targeting **BDNF V66M variant:** insights from in silico docking and molecular analysis. Amb Express. 2023 Nov 28;13(1):134.

5. [2023]

Sakhawat A, Awan SJ, Khan MU, Shahid S, Maqbool T, Zubair HM, Manzoor H, Khan S. In silico and in vitro analyses to investigate the effects of vitamin C on **VEGF protein**. Journal of Taibah University Medical Sciences. 2024 Aug 1;19(4):775-89.

Open Exploration Journal

2024-till now

PloseOne

SKILLS & COMPETENCIES Technical & Laboratory Skills

- Mammalian cell culture
- Stem cell culture & Plant-derived **exosome isolation and characterization**
- Animal models for drug evaluation
- Protein encapsulation & delivery systems
(*Exosomes – mammalian, stem cell & plant-derived* | *PLGA nanoparticles*)
- Molecular cloning
- PCR | ELISA | Western blot | 2D electrophoresis
- Industrial-scale fermentation optimization
- Ultrafiltration & diafiltration (biopharmaceutical formulation)
- Bioprocess column packing (lab to production scale) □ Pilot-scale therapeutic protein production (clinical studies)
- Bioprocess & analytical equipment handling:
HPLC | ÄKTA Pilot | ÄKTA Explorer | Fermentation systems | Uniflux 30 | Constant System Cell Disrupter | New Brunswick Bioreactor
- Protein purification, characterization & QC

Computational Biology & Bioinformatics

- Genetic data analysis (UCSC, NCBI, Ensembl)
- Primer design (Primer3, Primer-BLAST, In-silico PCR)
- Protein structure prediction (Robetta, I-TASSER, SWISS-MODEL)
- ADMET & pharmacokinetic profiling
- Protein structure validation
- Protein–protein interaction analysis
- Nano informatics
- Immunoinformatic
- Mutational analysis
- Drug designing & Molecular docking (Phytocompounds, peptides, fusion proteins, Vaccine design)

- Docking visualization & interaction analysis (PyMOL, Chimera)
- Molecular dynamics simulations

Programming, Data Science & AI

- **Python:** Data analysis, scripting, ML
 - **R Programming:** Statistical analysis, visualization, R Shiny app development
 - Cheminformatics & QSAR modelling
 - Molecular descriptor generation & predictive modelling
 - **ML-based** phytochemical classification
 - PCA, clustering, ROC-AUC & confusion matrix evaluation
- ## Soft & Professional Skills
- Precision & manual dexterity
 - Critical analysis & problem-solving
 - Rapid experiential learner
 - Adaptability in multidisciplinary environments
 - Scientific communication & teamwork
 - Project planning & execution
 - Accountability & regulatory awareness

EDUCATION

PhD in Biochemistry---CGPA: 3.8

The University of Lahore, Pakistan

- **Thesis Title:** *Encapsulation of Selected Venom Peptides with Biodegradable Nanoparticles for the Treatment of Breast Cancer*
- **Research papers published 2025**

M.Phil. Biochemistry — CGPA: 3.75

The University of Lahore, Pakistan

- **Thesis Title:** *Effects of Vitamin C on Cancer cell proliferation via de-simulation of proliferative genes.*
- **Research papers published 2023**

Master of Science (MSc) in Biochemistry — CGPA: 3.55

The University of Lahore, Pakistan

- **Thesis Title:** *Effect of Cinnamomum zeylanicum (Ceylon Cinnamon) Aqueous Extract on Biochemical Parameters and Liver Histopathology in Alloxan-Induced Diabetic Rats.*

WORKSHOPS/SEMINAR/CONFERENCES ORGANIZED & PARTICIPATIONS

- **Therapeutic Applications of Stem Cells**, The University of Lahore – 5th January 2019
- **Symposium on Cells and Cancer Biology**, The University of Lahore – 14th March 2019
- **1st & 3rd National Conference on Bioactivity of Phytochemicals**, The University of Lahore – 26th–27th November 2019
- **R-Programming Workshop**, University of Management and Technology – 2023
- **Bioinformatics Training Workshop I for Wet Lab Life Sciences**, The University of Lahore – 2024

RECOMMENDATION & REFERENCES

➤ **Name: DR. NADEEM AHMED | Ph.D. Molecular Biology | Associate Professor | Head**

Biopharmaceuticals and Proteomics (BPL), CEMB

I strongly recommend Azra Sakhawat as a talented and dedicated biochemist. Her expertise in **molecular biology, microbiology, Exosomes isolation, bioinformatics, protein expression, biosimilars, and drug studies** has greatly contributed to our research at CEMB. She is hardworking, innovative, and will be a valuable asset to any scientific team.

Email: nadeem.ahmed@cemb.edu.pk | **Phone number:** (+92) 3014727440

➤ **Name: DR. SAAD TAHIR | Pharm D | Ph.D. Molecular Biology | Research Officer | Biopharmaceuticals and Proteomics (BPL), CEMB**

I strongly endorse Azra Sakhawat for her exceptional skills in biochemistry. Her contributions in **molecular biology, biochemistry, microbiology, Exosomes isolation, bioinformatics & biosimilars** have been invaluable to our work at CEMB. Her commitment and innovative mindset will benefit any team. **Email:** saad.cemb@yahoo.com | **Phone number:** (+92) 3314452557

Driven and enthusiastic, I am eager to leverage my expertise in biochemistry, molecular biology, and therapeutics to contribute to groundbreaking research, develop innovative solutions, and grow as a dynamic scientist in a progressive organization